

Chemistry Ch 2.4: Solutions and Solubility

Chapter 2, Matter · kitab ke page 15 se 26 · sections 2.4 aur 2.4.7 se 2.4.8

Solution

Ek **homogeneous** mixture jis ke zarray **1 nanometre se chhote** hon aur jo **mustahkam (stable)** ho.

Solute: jo ghulta hai. **Solvent:** jo ghol raha hai.

Aqueous solution: jo paani mein ghula ho. Paani ko **universal solvent** kaha jata hai.

Qism	Misalein
Gaseous	Haber's process mein N_2 aur H_2 , dhund (fog), dhuan
Liquid	Carbonated drinks, sirka, brine (namkeen paani)
Solid	Nickel par hydrogen, amalgam, alloys

Teen halatein, ye har baar poochhi jati hain

Unsaturated: zyadatar se **kam** solute. Aur ghol sakta hai.

Saturated: us temperature par **zyada se zyada** solute ghul chuka. Ab **dynamic equilibrium** hai.

Supersaturated: saturated se bhi **zyada**. Ye **ghair mustahkam (unstable)** hai, aur banti hai pehle **garam** kar ke phir **thanda** kar ke.

Alag baat: **concentrated aur dilute** sirf solute ki **nisbatan miqdar** batate hain. Brine concentrated hai.

Solubility aur temperature

Solubility: kisi khaas solvent mein, kisi khaas temperature par, zyada se zyada kitna solute ghul sakta hai.

Zyadatar solute: temperature barhne se solubility **barhti** hai. KCl 34.7 se 56 g, NH_4Cl 37.5 se 77 g.

Kuch ki kam hoti hai: $Ca(OH)_2$ 0.173 se **0.066** g, aur Na_2SO_4 .

Ye do istisna yaad rakhein, sawal inhi par banta hai.

Quiz

1. Define **solution**, **solute** and **solvent**, then name the three types of solution with one example of each.
2. Explain the difference between **saturated**, **unsaturated** and **supersaturated** solutions, and name two substances whose solubility **decreases** as temperature rises.